Q1. JEE Main 2026 (23 January Shift 1)

Let α and β respectively be the maximum and the minimum values of the function
$f(\theta)=4\left(\sin^4\left(\frac{7\pi}{2}-\theta\right)+\sin^4(11\pi+\theta)\right)-2\left(\sin^6\left(\frac{3\pi}{2}-\theta\right)+\sin^6(9\pi-\theta)\right), \theta \in \mathbf{R}$. Then $\alpha+2\beta$ is equal to :

(1) 4 (2) 6 (3) 5 (4) 3

Q2. JEE Main 2026 (22 January Shift 2)

Let $\cos(\alpha+\beta)=-\frac{1}{10}$ and $\sin(\alpha-\beta)=\frac{3}{8}$, where $0<\alpha<\frac{\pi}{3}$ and $0<\beta<\frac{\pi}{4}$. If $\tan 2\alpha=\frac{3(1-r\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}, r, s \in N$, then $r+s$ is equal to ____ .

Q3. JEE Main 2026 (22 January Shift 1)

If $\frac{\cos^2 48^\circ-\sin^2 12^\circ}{\sin^2 24^\circ-\sin^2 6^\circ}=\frac{\alpha+\beta\sqrt{5}}{2}$, where $\alpha, \beta \in \mathbb{N}$, then $\alpha+\beta$ is equal to _____

Q4. JEE Main 2026 (23 January Shift 2)

The least value of $\left(\cos^2\theta-6\sin\theta\cos\theta+3\sin^2\theta+2\right)$ is

(1) -1 (2) 1 (3) $4-\sqrt{10}$ (4) $4+\sqrt{10}$

Q5. JEE Main 2026 (23 January Shift 2)

Let $\frac{\pi}{2}<\theta<\pi$ and $\cot\theta=-\frac{1}{2\sqrt{2}}$. Then the value of
$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta+\sin 8\theta)+\cos\left(\frac{15\theta}{2}\right)(\cos 8\theta-\sin 8\theta)$
is equal to

(1) $-\frac{\sqrt{2}}{\sqrt{3}}$ (2) $\frac{\sqrt{2}-1}{\sqrt{3}}$ (3) $\frac{\sqrt{2}}{\sqrt{3}}$ (4) $\frac{1-\sqrt{2}}{\sqrt{3}}$

Q6. JEE Main 2026 (21 January Shift 1)

The value of $\operatorname{cosec} 10^\circ-\sqrt{3}\sec 10^\circ$ is equal to :

(1) 8 (2) 6 (3) 2 (4) 4

Q7. JEE Main 2026 (28 January Shift 1)

If $\frac{\tan(\mathrm{A}-\mathrm{B})}{\tan \mathrm{A}}+\frac{\sin^2 \mathrm{C}}{\sin^2 \mathrm{A}}=1,\ \mathrm{A},\ \mathrm{B}, \mathrm{C} \in \left(0, \frac{\pi}{2}\right)$, then

(1) $\tan \mathrm{A}, \tan \mathrm{C}, \tan \mathrm{B}$ are in A.P.

(2) $\tan A, \tan B, \tan C$ are in G.P.

(3) $\tan A, \tan C, \tan B$ are in G.P.

(4) $\tan \mathrm{A}, \tan \mathrm{B}, \tan \mathrm{C}$ are in A.P.

Q8. JEE Main 2026 (24 January Shift 1)

If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then $\sin 7x\left(\cos\frac{13x}{2} + \sin\frac{13x}{2}\right) + \cos 7x\left(\cos\frac{13x}{2} - \sin\frac{13x}{2}\right)$ is equal to

(1) $\frac{4}{\sqrt{26}}$ (2) $\frac{6}{\sqrt{26}}$ (3) $\frac{5}{\sqrt{13}}$ (4) $\frac{1}{\sqrt{13}}$

Q9. JEE Main 2026 (24 January Shift 1)

The value of $\frac{\sqrt{3}\operatorname{cosec} 20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to

(1) 16 (2) 32 (3) 64 (4) 12

ANSWERS AND SOLUTIONS

1. (3) **2.** 20 **3.** 4 **4.** (3) **5.** (4) **6.** (4) **7.** (3) **8.** (4)
9. (3)

1. **(3)** $f(\theta) = 4\left\{\sin^4\left(\frac{7\pi}{2} - \theta\right) + \sin^4(11\pi + \theta)\right\} - 2\left\{\sin^2\left(\frac{3\pi}{2} - \theta\right) + \sin^2(9\pi - \theta)\right\}$

$f(\theta) = 4(\cos^4(\theta) + \sin^4(\theta)) - 2(\cos^6\theta + \sin^6\theta)$

$f(\theta) = 4(1 - 2\sin^2\theta\cos^2\theta) - 2(1 - 3\sin^2\theta\cos^2\theta)$

$f(\theta) = 2 - 2\sin^2\theta\cos^2\theta$

$f(\theta) = 2 - \frac{\sin^2(2\theta)}{2}$

$\alpha = f(\theta)_{max} = 2$

$\beta = f(\theta)_{min} = \frac{3}{2}$

$\Rightarrow \alpha + 2\beta = 5$

2. **(20)** $\tan 2\alpha = \tan[(\alpha+\beta) + (\alpha-\beta)]$

$\tan 2\alpha = \frac{\tan(\alpha+\beta)+\tan(\alpha-\beta)}{1-\tan(\alpha+\beta)\cdot\tan(\alpha-\beta)}$

$\tan 2\alpha = \frac{\left(-\sqrt{99}+\frac{3}{\sqrt{55}}\right)}{1-(\sqrt{99})\left(\frac{3}{\sqrt{55}}\right)}$

$\tan 2\alpha = \frac{-3\sqrt{11}+\frac{3}{\sqrt{5}\times\sqrt{11}}}{1+\frac{9\sqrt{11}}{\sqrt{5}\times\sqrt{11}}}$

$\tan 2\alpha = \frac{3(1-11\sqrt{5})}{\sqrt{11}(9+\sqrt{5})}$

$\mathrm{r} = 11,\ \mathrm{s} = 9$

$\mathrm{r} + \mathrm{s} = 20$

3. **(4)** Using $\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)$:

$\cos^2 48^\circ - \sin^2 12^\circ = \cos 60^\circ \cos 36^\circ = \frac{1}{2}\cos 36^\circ$.

Using $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$:

$\sin^2 24^\circ - \sin^2 6^\circ = \sin 30^\circ \sin 18^\circ = \frac{1}{2}\sin 18^\circ$.

Ratio $= \frac{\cos 36^\circ}{\sin 18^\circ} = \frac{(1+\sqrt{5})/4}{(\sqrt{5}-1)/4} = \frac{(1+\sqrt{5})^2}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$.

$\alpha = 3,\ \beta = 1.\ \alpha + \beta = 4.$

4. **(3)** $\cos^2\theta + 3\sin^2\theta - 3\sin 2\theta + 2.$

Using double angle formulas:

$= \frac{1+\cos 2\theta}{2} + \frac{3(1-\cos 2\theta)}{2} - 3\sin 2\theta + 2$

$= 4 - \cos 2\theta - 3\sin 2\theta.$

Minimum of $-\cos 2\theta - 3\sin 2\theta = -\sqrt{1+9} = -\sqrt{10}$.

Least value $= 4 - \sqrt{10}$.

5. **(4)** Expanding and regrouping:

$\sin\frac{15\theta}{2}(\cos 8\theta + \sin 8\theta) + \cos\frac{15\theta}{2}(\cos 8\theta - \sin 8\theta)$

$= \left(\cos\frac{15\theta}{2}\cos 8\theta + \sin\frac{15\theta}{2}\sin 8\theta\right) + \left(\sin\frac{15\theta}{2}\cos 8\theta - \cos\frac{15\theta}{2}\sin 8\theta\right)$

$= \cos\left(8\theta - \frac{15\theta}{2}\right) + \sin\left(\frac{15\theta}{2} - 8\theta\right) = \cos\frac{\theta}{2} - \sin\frac{\theta}{2}$.

From $\cot\theta = -\frac{1}{2\sqrt{2}}$: $\sin\theta = \frac{2\sqrt{2}}{3}$, $\cos\theta = -\frac{1}{3}$.

$\left(\cos\frac{\theta}{2} - \sin\frac{\theta}{2}\right)^2 = 1 - \sin\theta = 1 - \frac{2\sqrt{2}}{3} = \frac{(\sqrt{2}-1)^2}{3}$.

Since $\frac{\pi}{4} < \frac{\theta}{2} < \frac{\pi}{2}$, $\sin\frac{\theta}{2} > \cos\frac{\theta}{2}$, so the value is $\frac{1-\sqrt{2}}{\sqrt{3}}$.

6. **(4)** $\operatorname{cosec} 10^\circ - \sqrt{3}\sec 10^\circ = \frac{\cos 10^\circ - \sqrt{3}\sin 10^\circ}{\sin 10^\circ \cos 10^\circ}$.

Numerator: $\cos 10^\circ - \sqrt{3}\sin 10^\circ = 2\left(\frac{1}{2}\cos 10^\circ - \frac{\sqrt{3}}{2}\sin 10^\circ\right) = 2\cos 70^\circ = 2\sin 20^\circ$.

Denominator: $\sin 10^\circ \cos 10^\circ = \frac{1}{2}\sin 20^\circ$.

Result $= \frac{2\sin 20^\circ}{\frac{1}{2}\sin 20^\circ} = 4$.

7. **(3)** Starting with $\frac{\tan(A-B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1$, rearrange to $\frac{\tan(A-B)}{\tan A} = 1 - \frac{\sin^2 C}{\sin^2 A}$.

$\Rightarrow \tan(A-B)\tan A = \sin^2 A - \sin^2 C$.

Applying the identity $\sin^2\theta = \frac{\tan^2\theta}{1+\tan^2\theta}$ and simplifying leads to a relationship between the tangent values.

The constraint reduces to $(\tan C)^2 = \tan A \cdot \tan B$, which means $\tan A$, $\tan C$, and $\tan B$ form a geometric progression with common ratio $r = \frac{\tan C}{\tan A}$.

8. **(4)** $\cot x = \dfrac{5}{12} \Rightarrow \cos x = \dfrac{-5}{13} = 2\cos^2\dfrac{x}{2} - 1$

$\cos\left(\dfrac{x}{2}\right) = -\dfrac{2}{\sqrt{13}}$ or $\dfrac{2}{\sqrt{13}}$ (rejected)

$\left\{\because \dfrac{x}{2} \in \left(\dfrac{\pi}{2}, \dfrac{3\pi}{4}\right)\right\}$

$\left(\sin 7x\dfrac{\sin 13x}{2} + \cos 7x\dfrac{\cos 13x}{2}\right) + \left(\sin 7x\dfrac{\cos 13x}{2} - \cos 7x\dfrac{\sin 13x}{2}\right)$

$\cos\left(7x - \dfrac{13x}{2}\right) + \sin\left(7x - \dfrac{13x}{2}\right)$

$\cos\dfrac{x}{2} + \sin\left(\dfrac{x}{2}\right)$

$\dfrac{3}{\sqrt{13}} - \dfrac{2}{\sqrt{13}} = \dfrac{1}{\sqrt{13}}$

9. **(3)** Numerator:

$\sqrt{3}\operatorname{cosec} 20^\circ - \sec 20^\circ = \frac{\sqrt{3}\cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cos 20^\circ}$

$\sqrt{3}\cos 20^\circ - \sin 20^\circ = 2\left(\frac{\sqrt{3}}{2}\cos 20^\circ - \frac{1}{2}\sin 20^\circ\right)$

$= 2\cos(20^\circ + 30^\circ) = 2\cos 50^\circ$

So numerator $= \frac{2\cos 50^\circ}{\sin 20^\circ \cos 20^\circ} = \frac{4\cos 50^\circ}{\sin 40^\circ} = 4$

Denominator:

$\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$ where $\cos 60^\circ = \frac{1}{2}$

Using product formula:

$\sin 20^\circ \cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}\sin 20^\circ$

Therefore $\cos 20^\circ \cos 40^\circ \cos 80^\circ \cdot \frac{1}{2} = \frac{1}{16}$

So, $\frac{4 \cdot \sin 20^\circ \cos 20^\circ}{\frac{1}{16}} = 64$